A
Real-Time Research Project
On

SAFTEY AND SMART HELMET SYSTEM FOR BIKE RIDERS

(Submitted in partial fulfillment of the requirements for the award of Degree)

BACHELOR OF TECHNOLOGY
In
COMPUTER SCIENCE AND ENGINEERING
By

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April, 2026.

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CERTIFICATE

This is to certify that the project entitled "**SAFTEY AND SMART HELMET SYSTEM FOR BIKE RIDERS**" being submitted by **A.ARAVIND(247R1A05CK)** in partial fulfillment of the requirements for the award of the degree of B.Tech in Computer Science and Engineering to the Jawaharlal Nehru Technological University Hyderabad, during the year 2025-26.

The results embodied in this have not been submitted to any other University or Institute for the award of any degree or diploma.

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A.ARAVIND (247R1A05CK)

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ABSTRACT

Road accidents are one of the major causes of injuries and fatalities worldwide. Many accidents occur due to the negligence of riders, such as not wearing helmets, drunk driving, or the delay in providing medical assistance after an accident. To address these issues, this project presents a **"Smart and Safety Helmet System for Bike Riders"**, an IoT-based safety solution designed to improve rider protection and enable rapid emergency response.

The proposed system integrates multiple sensors and communication modules within a smart helmet to enhance road safety.. A **helmet detection mechanism** ensures that the rider wears the helmet before the ignition system is activated. Additionally, an **accelerometer or MEMS sensor** is used to detect sudden impacts or accidents.

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1. INTRODUCTION

1.INTRODUCTION

The project titled **"Smart and Safety Helmet System for Bike Riders"** is designed to develop an intelligent safety system that enhances the protection of two-wheeler riders using embedded systems and IoT technologies. The system aims to reduce road accidents and improve emergency response by integrating sensors and communication modules into a smart safety mechanism.

Road accidents involving two-wheelers have increased significantly due to negligence such as not wearing helmets, drunk driving, overspeeding, and delayed medical assistance after accidents. According to traffic safety reports, many fatalities occur because riders either fail to wear helmets or do not receive timely help after an accident.

To address these problems, the proposed system introduces a smart helmet safety solution that ensures rider safety by monitoring important conditions. The system uses sensors such as MEMS (Micro Electro Mechanical System) sensors for accident detection and a GSM module (SIM800/SIM900) to send emergency alerts. When an accident occurs, the system automatically sends a notification message to predefined emergency contacts with the rider's location information.

The system also includes components such as NodeMCU (ESP8266), LCD display, buzzer, and power supply, which work together to monitor the rider's condition and provide alerts when necessary. By integrating these hardware modules, the system creates a reliable mechanism for improving road safety and reducing accident-related risks.

This project demonstrates how embedded systems and IoT technology can be effectively applied to develop intelligent transportation safety solutions that protect riders and ensure quick emergency response.

1.2 PROJECT PURPOSE

The primary purpose of this project is to enhance the safety of bike riders and reduce road accidents by developing an intelligent helmet monitoring system.

The purpose of the Smart and Safety Helmet System is to:

• Improve the safety of motorcycle riders

• Detect accidents automatically using MEMS sensors

• Send emergency alert messages to predefined contacts.

1.3 PROJECT FEATURES

The Smart and Safety Helmet System for Bike Riders includes several functional and technical features that improve rider safety and enable quick emergency response during accidents.

1. Accident Detection using MEMS Sensor

The system uses a MEMS (Micro Electro Mechanical System) sensor to detect sudden impacts, vibrations, or abnormal movements that indicate a possible accident. When the sensor detects a strong impact beyond the predefined threshold, the system identifies it as an accident and triggers the alert mechanism.

2. Automatic Emergency Alert System

In the event of an accident, the system automatically sends an emergency alert message to predefined contact numbers using the GSM module (SIM800/SIM900). This message informs family members or emergency services about the accident so that immediate help can be provided.

3. GSM-Based Communication

The GSM module enables wireless communication by sending SMS alerts. Once an accident is detected, the system uses the inserted SIM card to transmit a message to registered phone numbers. This ensures that emergency notifications are delivered quickly.

4. Real-Time Monitoring

The system continuously monitors the rider's movement and detects abnormal conditions using sensors.

5. LCD Display for System Status

An **LCD display** is used to show important information such as system status, alerts, and sensor readings. It helps the user understand whether the system is functioning correctly.

6. Buzzer Alert Mechanism

A buzzer is included in the system to provide an audible alert during emergency situations. When an accident is detected or a warning condition occurs, the buzzer produces a sound signal to indicate the alert.

7. Power Supply Module

The system operates using a regulated power supply unit that provides stable voltage to all components such as the NodeMCU, sensors, GSM module, and LCD display.

2.LITERATURE SURVEY

2.1 LITERATURE SURVEY

The field of review analysis and opinion mining has evolved significantly with the advancement of The rapid increase in road traffic and the number of two-wheeler vehicles has resulted in a significant rise in road accidents worldwide. Motorcyclists are among the most vulnerable road users because they have minimal physical protection compared to other vehicle drivers. According to traffic safety reports, a large percentage of fatalities occur due to riders not wearing helmets, driving under the influence of alcohol, or delays in receiving emergency medical assistance after accidents. These issues have motivated researchers and engineers to develop intelligent safety systems that can help prevent accidents and provide immediate assistance when they occur.

In the early stages of road safety systems, the primary focus was on passive safety mechanisms, such as helmets, protective jackets, and seat belts. Helmets are one of the most effective protective devices for motorcyclists, reducing the risk of severe head injuries during accidents. However, traditional helmets only provide physical protection and do not have the capability to monitor the rider's condition or send alerts during emergencies. As a result, many accident victims remain unattended for long periods, especially when accidents occur in remote areas.

To address these limitations, researchers began exploring electronic safety systems integrated with helmets and motorcycles. One of the earliest developments in this field was the introduction of helmet detection systems. These systems ensured that the motorcycle engine would only start if the rider was wearing a helmet. This approach encouraged riders to follow safety regulations and helped reduce injuries caused by riding without helmets. However, these systems mainly focused on helmet usage and did not provide accident detection or emergency communication features.

With the advancement of embedded systems and sensor technology, researchers began integrating various sensors into helmets and vehicles to monitor rider safety. Sensors such as accelerometers and MEMS (Micro Electro Mechanical System) sensors were widely used to detect sudden impacts, vibrations, or abnormal movements that indicate accidents. MEMS sensors are compact, lightweight, and highly sensitive, making them suitable for detecting rapid changes in motion. When a sudden shock or unusual tilt is detected, the system can interpret it as an accident and trigger a response mechanism.

Another significant development in smart helmet systems is the use of communication technologies such as GSM (Global System for Mobile Communication). GSM modules like SIM800 and SIM900 enable devices to send SMS messages or make calls using mobile networks. By integrating GSM modules into accident detection systems, researchers developed solutions that automatically send emergency alert messages to predefined contacts such as family members, friends, or emergency services. This feature ensures that help can reach the accident victim as quickly as possible.

In addition to accident detection, researchers have proposed systems that focus on

preventing accidents caused by drunk driving. Alcohol detection sensors such as the MQ-3 sensor are commonly used to measure the alcohol concentration in a rider's breath. When alcohol levels exceed a predefined threshold, the system prevents the motorcycle engine from starting. This mechanism discourages riders from driving under the influence of alcohol and helps reduce accident rates.

Recent research has focused on combining multiple safety features into a single integrated smart helmet system. These systems include accident detection, alcohol detection, helmet usage monitoring, and emergency communication in one platform. Microcontrollers such as Arduino or NodeMCU (ESP8266) are commonly used to control these systems because they provide efficient processing capabilities and support communication with sensors and modules.

Despite the significant progress in smart helmet technologies, several challenges still exist. Some systems require complex hardware configurations and high power consumption. Others depend heavily on internet connectivity or GPS modules, which may not always be reliable in rural or underground areas. Additionally, the cost of advanced systems may limit their accessibility for widespread use.

These systems ensured that the motorcycle engine would only start if the rider was wearing a helmet. This approach encouraged riders to follow safety regulations and helped reduce injuries caused by riding without helmets. However, these systems mainly focused on helmet usage and did not provide accident detection or emergency communication features.

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The proposed Smart and Safety Helmet System for Bike Riders aims to address some of these challenges by designing a simple, cost-effective, and reliable accident detection and alert system. The system uses a MEMS sensor to detect accidents, a GSM module (SIM800/SIM900) to send emergency messages, and supporting components such as an LCD display, buzzer, and power supply to provide alerts and system information. By integrating these technologies, the system ensures quick accident detection and immediate communication with emergency contacts.

In conclusion, the development of smart helmet systems represents an important step toward improving road safety for motorcyclists. From basic helmet protection to advanced sensor-based monitoring and communication technologies, researchers have continuously worked to enhance rider safety. The proposed system builds upon these developments by providing a practical and efficient safety solution that helps reduce accident-related risks and ensures faster emergency response.

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2.2 REVIEW OF RELATED WORK

2.2.1 REVIEW OF RELATED WORK

The increasing number of road accidents involving motorcycle riders has become a major concern across the world. Two-wheelers provide convenience and fuel efficiency, but they also expose riders to higher risks compared to other vehicles. According to traffic safety studies, a large number of fatal accidents occur due to lack of safety measures, drunk driving, delayed medical response, and failure to wear protective helmets. Because of these issues, researchers have been working on intelligent safety systems that can monitor rider behavior and provide immediate assistance during emergencies.

Over the past decade, many researchers have proposed different approaches to improve rider safety using embedded systems, sensor technologies, and wireless communication modules. These systems focus on accident detection, helmet usage monitoring, alcohol detection, and automatic emergency alert systems. This section presents a detailed review of related work in the field of smart helmet systems and motorcycle safety technologies.

Initially, motorcycle safety research mainly focused on the use of protective helmets. Helmets were designed to reduce the severity of head injuries during accidents by absorbing impact forces. Several studies confirmed that wearing helmets significantly reduces the risk of fatal head injuries.

However, traditional helmets only provide physical protection and do not include any electronic monitoring features. Riders may still ignore safety rules such as wearing helmets properly or avoiding drunk driving. Because of these limitations, researchers began exploring the integration of electronics into helmets to improve rider safety.

2.2.2 Helmet Detection and Ignition Control Systems

One of the early smart helmet systems introduced a helmet detection mechanism that prevents the motorcycle from starting unless the rider wears the helmet. These systems used sensors or switches installed inside the helmet and connected to the motorcycle's ignition

system.

Research showed that this approach encourages riders to follow safety rules and reduces accidents caused by riding without helmets. However, these systems only focused on helmet usage and did not provide accident detection or emergency communication features.

2.2.4 Accident Detection Using Mems Sensors

Accident detection is another important area of research in motorcycle safety systems. Many researchers have used MEMS (Micro Electro Mechanical System) sensors and accelerometers to detect sudden movements or impacts during accidents.

MEMS sensors are capable of detecting rapid changes in motion, vibration, and orientation. When a collision occurs, the sensor readings change significantly, indicating an abnormal event.

However, accident detection alone is not sufficient unless it is combined with an emergency alert mechanism to notify others about the accident.

2.2.5 GSM-Based Emergency Alert Systems

To improve emergency response time, researchers introduced GSM communication modules into accident detection systems. GSM modules such as SIM800 or SIM900 allow embedded devices to send SMS messages through cellular networks.

When an accident is detected, the microcontroller activates the GSM module and sends an alert message to predefined contact numbers such as family members, friends, or emergency services.

Research findings indicate that GSM-based alert systems significantly reduce rescue time because they automatically notify others about the accident location and situation. This feature is especially useful when the rider is unconscious and unable to call for help.

2.1.6 Integrated Smart Helmet Systems

Recent research focuses on integrating multiple safety features into a single smart helmet system. These systems combine helmet detection, alcohol sensing, accident detection, communication modules, and alert mechanisms.

Microcontrollers such as Arduino and ESP32 are commonly used to control these components. These controllers process sensor data, detect abnormal conditions, and trigger appropriate responses.

2.2.7 Identified Research Gaps

Based on the review of related work, several research gaps have been identified:

1. Some systems focus only on helmet detection without accident monitoring.
2. Many accident detection systems lack automatic communication features.
3. Some solutions require complex hardware configurations and high power consumption.
4. A few systems integrate alcohol detection but do not combine it with accident alerts.

5.Many existing implementations are expensive and difficult to deploy for regular riders.

These limitations highlight the need for a comprehensive and efficient smart helmet system.

2.2.8 Justification of the Proposed System

The proposed **Safety and Smart Helmet System for Bike Riders** aims to overcome the limitations of existing systems by integrating multiple safety features into a single embedded system.The system uses MEMS sensors for accident detection and GSM communication for sending emergency alerts. Additional components such as LCD display and buzzer provide real-time system status and alert notifications.

This approach ensures:

• Improved rider safety
• Automatic accident detection
• Quick emergency notification
• Low cost and simple implementation
• Reliable operation using embedded technology

By combining sensor technology with wireless communication, the proposed system provides a practical and effective solution for enhancing motorcycle rider safety.

2.3 DEFINITION OF PROBLEM STATEMENT

Road accidents involving motorcycle riders are increasing every year, resulting in serious injuries and loss of lives. Many accidents occur due to drunk driving, failure to wear helmets, and delayed emergency response after accidents.

In many cases, accident victims are unable to contact emergency services because they are unconscious or severely injured. As a result, medical assistance is delayed, which increases the risk of fatalities.

These systems ensured that the motorcycle engine would only start if the rider was wearing a helmet.This approach encouraged riders to follow safety regulations and helped reduce injuries caused by riding without helmets. However, these systems mainly focused on helmet usage and did not provide accident detection or emergency communication features.

With the advancement of embedded systems and sensor technology, researchers began integrating various sensors into helmets and vehicles to monitor rider safety. Sensors such as accelerometers and MEMS (Micro Electro Mechanical System) sensors were widely used to detect sudden impacts, vibrations, or abnormal movements that indicate accidents. MEMS sensors are compact, lightweight, and highly sensitive, making them suitable for detecting rapid changes in motion. When a sudden shock or unusual tilt is detected, the system can interpret it as an accident and trigger a response mechanism

The system uses a MEMS sensor to detect sudden movements or impacts during accidents. When an accident occurs, the microcontroller processes the sensor data and triggers the emergency alert system.

A GSM module is used to send an SMS alert message to predefined emergency contacts. This message informs them about the accident so that immediate help can be provided.

This integrated approach improves safety by combining accident detection and communication technology in a single compact system.

Therefore, the problem is to develop a smart helmet system that can monitor rider safety, detect accidents using sensors, and automatically send emergency alerts through communication modules.

The proposed system aims to solve this problem by integrating MEMS sensors and GSM technology to create a smart helmet capable of detecting accidents and notifying emergency contacts instantly.

2.4 EXISTING SYSTEM

In the existing system, motorcycle safety mainly relies on traditional helmets and manual monitoring. Riders are responsible for following safety rules such as wearing helmets and avoiding drunk driving.

. However, these systems mainly focused on helmet usage and did not provide accident detection or emergency communication features.

With the advancement of embedded systems and sensor technology, researchers began integrating various sensors into helmets and vehicles to monitor rider safety. Sensors such as accelerometers and MEMS (Micro Electro Mechanical System) sensors were widely used to detect sudden impacts, vibrations, or abnormal movements that indicate accidents. MEMS sensors are compact, lightweight, and highly sensitive, making them suitable for detecting rapid changes in motion. When a sudden shock or unusual tilt is detected, the system can interpret it as an accident and trigger a response mechanism

Some systems include helmet detection mechanisms that prevent the engine from starting if the helmet is not worn. Other systems focus on alcohol detection to prevent drunk driving.

Although these systems improve safety to some extent, they are limited in functionality. Many existing systems do not provide accident detection or emergency alert mechanisms.

As a result, accident victims may not receive immediate assistance, leading to serious consequences.

The system uses a MEMS sensor to detect sudden movements or impacts during accidents. When an accident occurs, the microcontroller processes the sensor data and triggers the emergency alert system.

A GSM module is used to send an SMS alert message to predefined emergency contacts. This message informs them about the accident so that immediate help can be provided.

The system also includes an LCD display to show system status and a buzzer to provide audible alerts.

This integrated approach improves safety by combining accident detection and communication technology in a single compact system.

Limitations of Existing System

The existing motorcycle safety systems have several limitations:

1. **Lack of Accident Detection**
   Many traditional systems cannot detect accidents automatically.
2. **Delayed Emergency Response**
   Victims may not receive help immediately after an accident.
3. **Limited Safety Features**
   Most systems focus on only one feature such as helmet detection or alcohol detection.
4. **Manual Monitoring**
   Safety depends mainly on rider responsibility rather than automated systems.
5. **No Communication System**
   Some systems do not include GSM or communication modules for sending alerts.
6. **Low Integration**
   Different safety features are not integrated into a single system.

Because of these limitations, there is a need for a smart and automated helmet system that can provide multiple safety functions.

2.5 PROPOSED SYSTEM

The proposed **Safety and Smart Helmet System for Bike Riders** is designed to improve rider safety by integrating sensor technology and communication modules.

The system uses a MEMS sensor to detect sudden movements or impacts during accidents. When an accident occurs, the microcontroller processes the sensor data and triggers the emergency alert system.

A GSM module is used to send an SMS alert message to predefined emergency contacts. This message informs them about the accident so that immediate help can be provided.

The system also includes an LCD display to show system status and a buzzer to provide audible alerts.

This integrated approach improves safety by combining accident detection and communication technology in a single compact system.

Advantages of the Proposed System

• Automatic accident detection using MEMS sensors
• Immediate emergency alert through GSM communication
• Improved rider safety and accident response time
• Low-cost and easy implementation
• Reliable embedded system design
• Real-time alerts using buzzer and LCD display

2.6 OBJECTIVES

The main objectives of the Smart Helmet System are:

1. To develop a smart helmet system that improves motorcycle rider safety.

2. To detect accidents using MEMS sensor technology.
3. To send automatic emergency alerts using GSM communication.
4. To provide real-time system status using LCD display and buzzer.
5. To reduce accident response time and improve rescue efficiency.
6. To design a cost-effective and reliable embedded safety system.

2.7 HARDWARE & SOFTWARE REQUIREMENTS

2.7.1 HARDWARE REQUIREMENTS

The hardware components required for the Smart Helmet System include:

• Arduino Microcontroller
• MEMS Sensor
• GSM Module
• LCD Display
• Buzzer
• Power Supply
• Connecting Wires

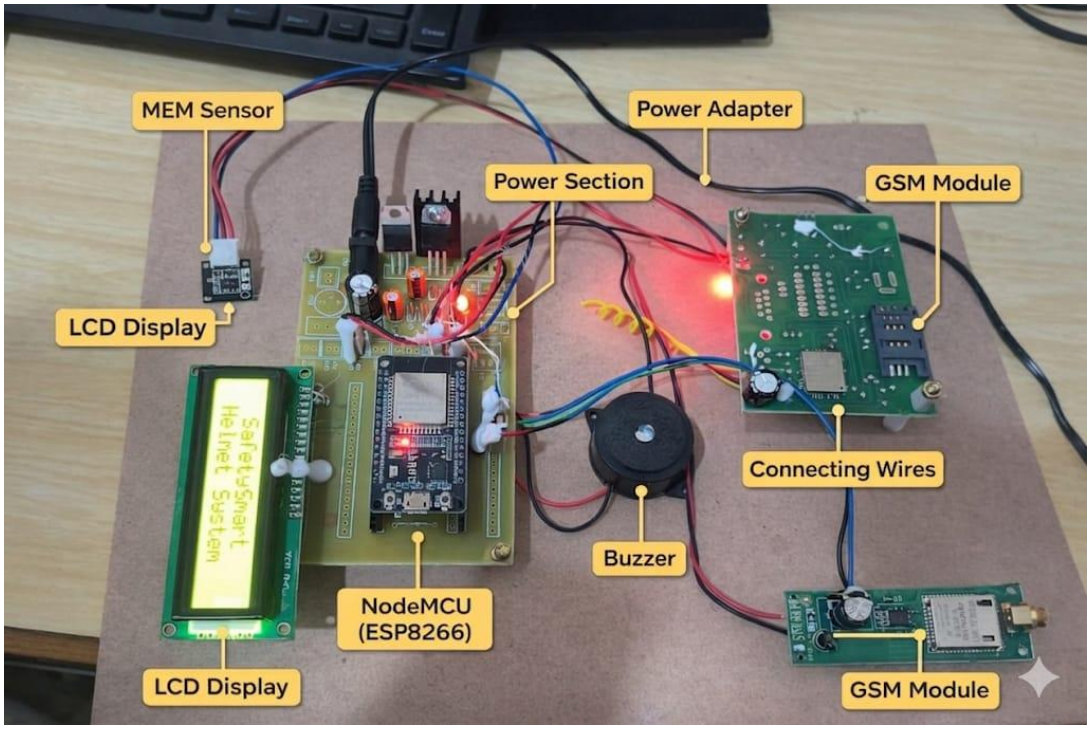

Fig 2.1 Hardware components

2.7.2 SOFTWARE REQUIREMENTS

The software tools required for the system are:

• Arduino IDE
• Embedded C Programming
• Proteus (for circuit simulation)
• Operating System: Windows 10 or above

3.SYSTEM ARCHITECTURE & DESIGN

3.1 SYSTEM ARCHITECTURE & DESIGN

The System Architecture of the Safety and Smart Helmet System for Bike Riders defines how different hardware and software components interact to monitor rider safety and respond to emergency situations. The system follows a structured process where sensors continuously monitor the rider and send signals to the microcontroller for processing**.**

This project architecture shows the procedure followed in the **Safety and Smart Helmet System**, starting from sensor detection to emergency alert generation

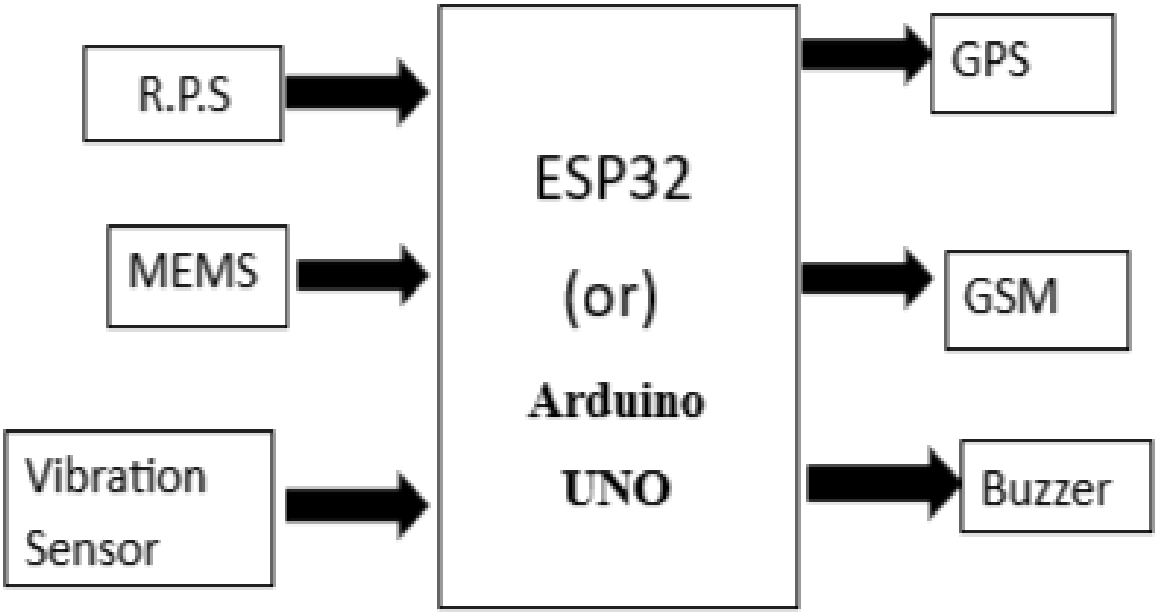

Figure 3.1: Project Architecture of Review Insights

3.2 DESCRIPTION

The project architecture represents the complete workflow of the **Safety and Smart Helmet System** from sensor monitoring to emergency alert generation.

The process begins with the **Sensor Monitoring Stage**, where the MEMS sensor continuously monitors the motion and orientation of the helmet. The sensor detects vibrations, sudden shocks, and unusual movements that may occur during an accident.

The sensor data is then sent to the **Microcontroller Processing Stage**. The microcontroller processes the incoming data and compares it with predefined threshold values to determine whether an accident has occurred. If the sensor readings exceed the threshold level, the system interprets the situation as a possible accident. The system then activates the **Alert Generation Stage**.

During this stage, the **GSM module** sends an SMS message to predefined emergency contacts such as family members or rescue services. This ensures that someone is immediately informed about the accident.Simultaneously, the **LCD display** provides visual information about the system status, and the **buzzer** produces an audible alarm to indicate an emergency situation.

This structured architecture ensures efficient accident detection, quick communication, and reliable operation of the system.

3.3 DATA FLOW DIAGRAM

A Data Flow Diagram (DFD) is a graphical representation that shows how data moves within a system. It illustrates how data enters the system, how it is processed, and how it is delivered as output.
In the Safety and Smart Helmet System, the DFD explains how sensor data is collected, processed by the microcontroller, and used to generate emergency alerts.

Levels of Data Flow Diagram
DFD is divided into different levels to represent the system in a structured and organized manner.

1. Level 0 DFD (Context Diagram)
Level 0 DFD provides a high-level overview of the entire system. It represents the system as a single process and shows its interaction with external entities.
In the Smart Helmet System:
External Entity: Rider / Emergency Contacts
Process: Smart Helmet Safety System
Input: Sensor Data (Motion / Impact Detection)
Output: Emergency Alert Message
Flow:
Rider → Smart Helmet System → Emergency Alert to Contacts

This level provides a simple overview of the system and its interaction with external users.

2. Level 1 DFD

Level 1 DFD breaks the main system into smaller sub-processes to provide more detailed information about system operations.

Main Processes:

1. Sensor Data Collection
2. Accident Detection Processing
3. Alert Generation
4. Communication through GSM Module
5. Notification to Emergency Contacts

Data Flow:

MEMS Sensor → Microcontroller
Microcontroller → Accident Detection Module
Accident Detection Module → GSM Module
GSM Module → Emergency Contacts
Microcontroller → LCD Display & Buzzer

This level provides a more detailed explanation of how the system processes data internally.

3. Level 2 DFD (Optional – Detailed Level)

Level 2 DFD further breaks down individual processes into smaller operations.

Example:

Accident Detection Process:

• Motion Data Monitoring
• Impact Detection
• Threshold Comparison
• Accident Confirmation

Alert Generation Process:

• Message Preparation
• GSM Module Activation
• SMS Transmission

This level provides a deeper understanding of the internal functioning of the system.

Benefits of Data Flow Diagram

1. **Clear Visualization** – Helps understand how data moves through the system.
2. **Improved System Design** – Makes system architecture structured and organized.
3. **Better Communication** – Easy to explain the system to examiners and reviewers.
4. **Error Identification** – Helps detect missing components or incorrect data flow.
5. **Modular Understanding** – Breaks complex systems into smaller manageable parts.

Applications of Data Flow Diagram

1. Used in system analysis and system design.
2. Helps in embedded system and software development planning.
3. Useful in representing hardware and software interactions.
4. Applied in IoT system architecture modeling.
5. Used in academic projects for explaining system functionality.

In the **Safety and Smart Helmet System**, the DFD explains how sensor data is processed and converted into emergency alerts through a structured sequence of operations

3.4 Data Flow Diagram

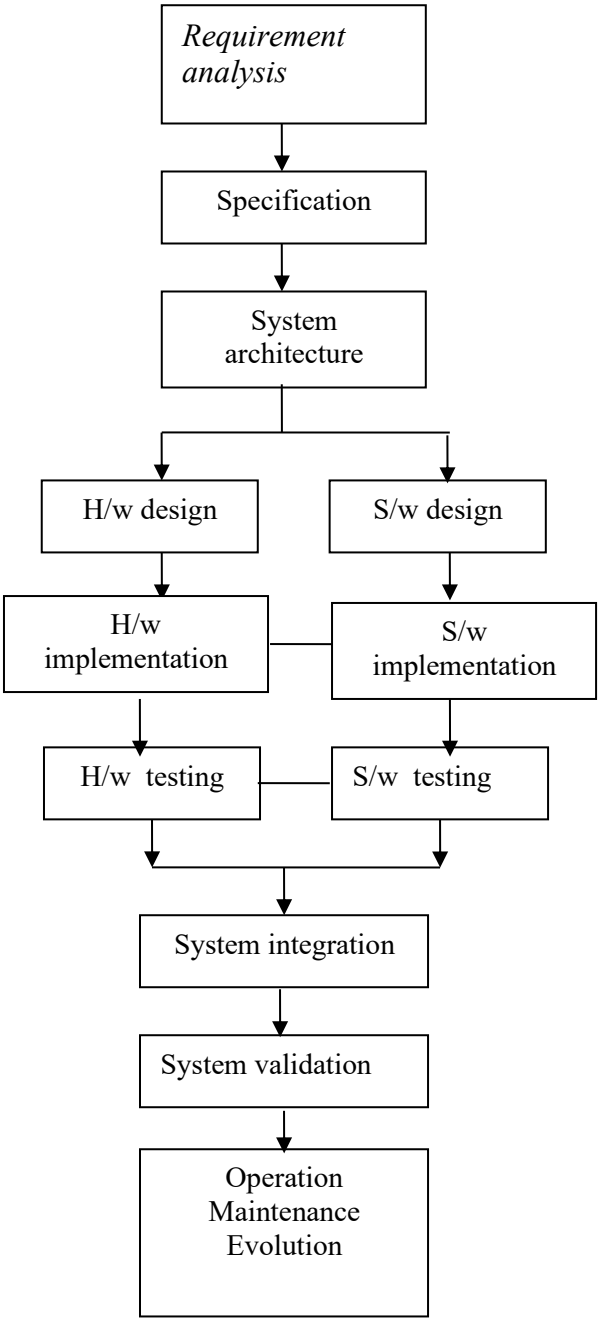

Figure 3.2: Data Flow Diagram of Safety and Smart Helmet System

4.IMPLEMENTATION

4.1 IMPLEMENTATION

The implementation phase of the Safety and Smart Helmet System for Bike Riders involves integrating hardware components, embedded programming, and communication technologies to create a reliable accident detection and emergency alert system. The primary objective of the implementation stage is to convert the designed architecture and system model into a working prototype that can monitor rider safety conditions in real time.

The system is implemented using an Arduino microcontroller, which acts as the central control unit of the entire system. Various sensors and modules such as the MEMS sensor, GSM communication module, LCD display, and buzzer are connected to the microcontroller to perform specific tasks. Each component works together to detect accidents and generate alerts automatically.

During implementation, the MEMS sensor continuously monitors the motion, orientation, and vibration of the helmet. These sensor readings are transmitted to the microcontroller at regular intervals. The microcontroller processes these signals and compares them with predefined threshold values. If the sensor values indicate sudden shock, abnormal tilt, or strong vibration, the system interprets it as an accident event.

Once the accident is detected, the system activates the emergency alert mechanism. The GSM module is used to send an SMS message to predefined emergency contacts such as family members, friends, or emergency services. This ensures that immediate help can be provided to the rider in case of an accident.

The LCD display module provides real-time system information such as system status, sensor monitoring status, and accident alert messages. This helps the user understand the working condition of the system.

The buzzer module is included to provide an audible alert when an accident is detected. The buzzer helps attract attention from nearby people and indicates that an emergency situation has occurred.

The implementation process includes the following stages:

1. Hardware component selection and integration
2. Circuit design and connection setup
3. Embedded programming using Arduino IDE
4. Sensor calibration and threshold configuration
5. GSM communication setup for SMS alerts
6. Testing and validation of accident detection logic

4.2 ALGORITHMS USED

The Smart Helmet System uses several algorithms and control procedures to monitor rider safety and detect accidents. These algorithms ensure efficient processing of sensor data and enable automatic emergency alerts.

1 Sensor Monitoring Algorithm

The sensor monitoring algorithm is responsible for continuously collecting motion data from the MEMS sensor installed inside the helmet. The MEMS sensor detects acceleration, tilt, and vibrations that occur during the movement of the rider.

The microcontroller periodically reads the sensor values and analyzes them to determine whether the motion is normal or abnormal.

Steps of Sensor Monitoring Algorithm

1. Initialize the MEMS sensor module.
2. Establish communication between the sensor and the microcontroller.
3. Continuously read sensor values from the MEMS module.
4. Convert the sensor signals into digital data.
5. Store and update sensor readings at regular intervals.
6. Send sensor data to the accident detection algorithm for further analysis.

Advantages

• Continuous monitoring of rider movement.
• Accurate detection of sudden vibrations and impacts.
• Low power consumption and efficient processing.

Disadvantages

• Sensor values may fluctuate due to road conditions.
• Requires proper calibration for accurate detection.

2. Accident Detection Algorithm

The accident detection algorithm analyzes the motion data obtained from the MEMS sensor to determine whether an accident has occurred. It compares the sensor readings with predefined threshold values to identify abnormal conditions.

When the vibration or tilt exceeds the threshold limit, the algorithm triggers the accident alert mechanism.

Steps of Accident Detection Algorithm

1. Receive sensor data from the monitoring module.
2. Compare sensor values with predefined threshold values.
3. Determine whether the motion is normal or abnormal.

4. If sensor values exceed the threshold limit:
   - Identify the event as a possible accident.
5. Trigger emergency alert system.
6. Activate buzzer and display alert message on LCD.
7. Send alert signal to GSM module.

Advantages
• Provides automatic accident detection.
• Reduces delay in emergency response.
• Simple and efficient implementation.
Disadvantages
• False alerts may occur in extremely rough road conditions.
• Requires careful tuning of threshold parameters.
3. **GSM Communication Algorithm**

Final Deployed Model

The selected model is integrated into a Streamlit-based web application. The workflow is:

User Review Input
↓
Text Preprocessing
↓
TF-IDF Feature Extraction
↓
Model Prediction
↓
Confidence Score
↓
Visualization

4.3 HARDWARE IMPLEMENTATION

The hardware implementation of the **Safety and Smart Helmet System for Bike Riders** involves assembling various electronic components and integrating them with the microcontroller to perform accident detection and emergency communication.

The hardware design focuses on creating a compact and reliable system that can be mounted inside the helmet without affecting the rider's comfort. Each hardware component plays an important role in ensuring proper system functionality.

The primary hardware components used in the system include:
• Arduino Microcontroller

• MEMS Sensor
• GSM Communication Module

• LCD Display
• Buzzer
• Power Supply Unit
• Connecting Wires and Circuit Board

The **Arduino microcontroller** acts as the central processing unit of the system. It receives input signals from the MEMS sensor and processes them to determine whether an accident has occurred. Based on the analysis, the microcontroller controls the GSM module, LCD display, and buzzer.

The **MEMS sensor** is responsible for detecting sudden movements, shocks, or abnormal vibrations. It continuously measures the motion of the helmet and sends data to the microcontroller for analysis.

The **GSM module** is used to send emergency alert messages to predefined contact numbers. It communicates with the microcontroller through serial communication and transmits SMS alerts when an accident is detected.

The **LCD display** provides visual feedback by displaying system status messages such as system initialization, monitoring status, and accident alerts.

The **buzzer module** generates an audible alarm during emergency conditions to alert nearby people.

All these components are connected through a circuit board using proper wiring connections. The power supply is provided through a battery source that powers the microcontroller and other modules.

This hardware setup ensures efficient operation of the Smart Helmet System and provides reliable accident detection and emergency alert functionality

4.4 SOFTWARE IMPLEMENTATION

The software implementation of the Smart Helmet System involves developing embedded code that controls the operation of sensors, communication modules, and alert mechanisms. The software is written using **Embedded C programming language** and developed using the **Arduino Integrated Development Environment (Arduino IDE)**. The program is uploaded to the Arduino microcontroller using a USB interface.

The software performs several tasks including:

1. Sensor data acquisition
2. Accident detection logic
3. Communication with GSM module
4. Display control for LCD module
5. Buzzer alert activation

The software continuously reads sensor values and processes them to determine whether an

accident has occurred. If the sensor values exceed predefined thresholds, the software triggers the emergency alert mechanism.
The GSM communication process is handled using **AT commands**, which are used to configure the GSM module and send SMS messages.

The software is designed to be efficient and lightweight so that it can run smoothly on the microcontroller without consuming excessive resources.

4.5 SYSTEM TESTING

System testing is an important stage in the implementation process. It ensures that all hardware and software components function correctly and that the system performs as expected under different conditions.
Testing of the Smart Helmet System includes the following steps:

1. Sensor Testing

The MEMS sensor is tested by generating controlled vibrations and movements to verify whether the sensor readings are detected correctly by the microcontroller.

2. GSM Module Testing

The GSM module is tested by sending sample SMS messages to predefined phone numbers. This ensures that the communication module works correctly and that the messages are transmitted successfully.

3. LCD Display Testing

The LCD display is tested by displaying various system messages such as startup messages, monitoring status, and accident alerts.

4. Buzzer Testing

The buzzer is tested by triggering the alert mechanism manually to confirm that it generates a proper audible alarm.

5. Integrated System Testing

After individual component testing, the entire system is tested as a complete unit. During testing, sudden motion or simulated accidents are introduced to verify whether the system detects the event and sends emergency alerts correctly.
Through these tests, the reliability and accuracy of the Smart Helmet System are verified.

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5.RESULTS &DISCUSSION

5.1 RESULTS & DISCUSSION

The following section presents the results obtained from the **Safety and Smart Helmet System for Bike Riders**. The results demonstrate how the system performs various safety functions such as accident detection, helmet monitoring, alert generation, and emergency communication.
The system was tested under different conditions to evaluate its performance and reliability. The experimental results confirm that the Smart Helmet System is capable of detecting accidents, generating alerts, and sending emergency notifications effectively.
The screenshots and experimental outputs illustrate the working behavior of the system components including the **microcontroller, MEMS sensor, GSM module, LCD display, and buzzer**.
These results validate the functionality of the proposed system and demonstrate its capability to enhance the safety of motorcycle riders.

5.1 System Initialization / Main Interface
In the below screen, the Smart Helmet System initializes all connected components when the power supply is turned on.
The LCD display shows the system startup message indicating that the helmet safety system is ready for operation.
During this stage, the microcontroller performs the following tasks:
• Initializes the Arduino microcontroller
• Establishes communication with the MEMS sensor
• Initializes the GSM communication module
• Activates the LCD display module
• Checks buzzer functionality
The LCD screen displays messages such as:
“SMART HELMET SYSTEM”
“SYSTEM READY”
This confirms that all system components are functioning correctly.
The system then enters the **monitoring mode**, where it continuously checks sensor values for abnormal motion or vibration.

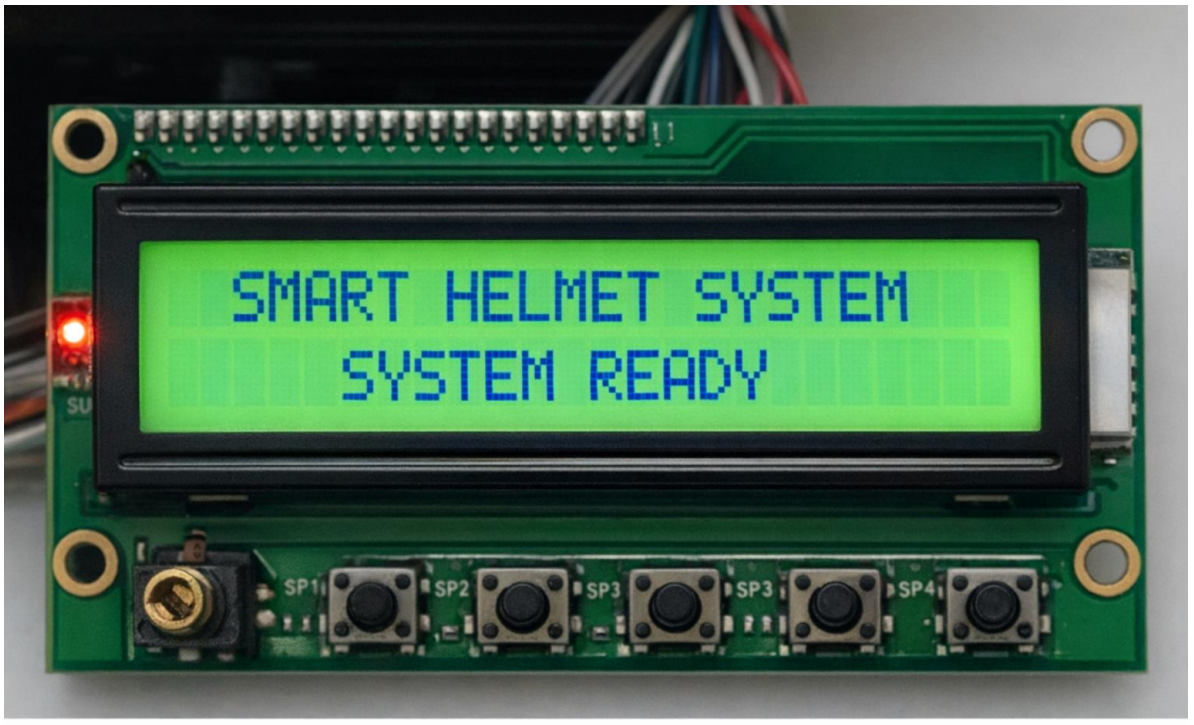


Figure 5.1: System Initialization of Smart Helmet System

Figure 5.1: System Initialization of Smart Helmet System

5.2 Sensor Monitoring Output

In the below screen, the system displays the sensor monitoring process.

The **MEMS sensor** continuously monitors the motion, vibration, and orientation of the helmet. The sensor sends real-time data to the microcontroller, which analyzes the values to determine whether the motion is normal or abnormal.

The sensor readings represent the movement of the helmet in different directions and help identify sudden shocks or unusual tilt conditions.

During normal riding conditions:

• Sensor values remain within predefined threshold limits

• The LCD display indicates **"Helmet Monitoring Active"**

• No emergency alerts are generated

This stage ensures continuous monitoring of the rider's safety condition.

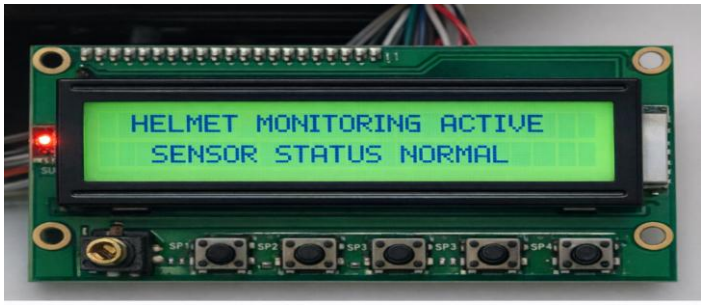


Figure 5.2: Sensor Monitoring Output

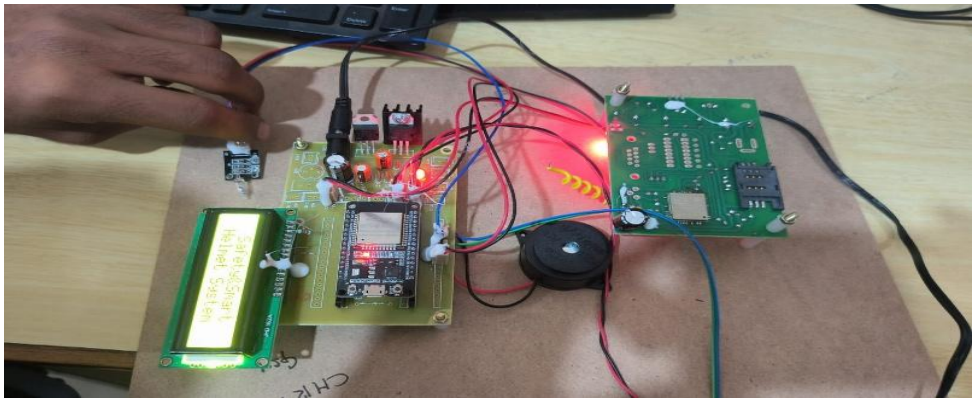


Figure 5.2: Sensor Monitoring Output

5.3 Accident Detection Result

In the below screen, the system detects an accident based on abnormal sensor readings.

When the MEMS sensor detects a sudden impact or excessive vibration exceeding the threshold value, the microcontroller interprets the event as a possible accident.

Once the accident is detected, the following actions are performed automatically:

• The microcontroller triggers the accident alert process

• The LCD screen displays **"ACCIDENT DETECTED"**

• The buzzer generates an audible alarm

• The GSM module prepares to send an emergency alert message

This process occurs instantly, ensuring that emergency alerts are generated without delay.

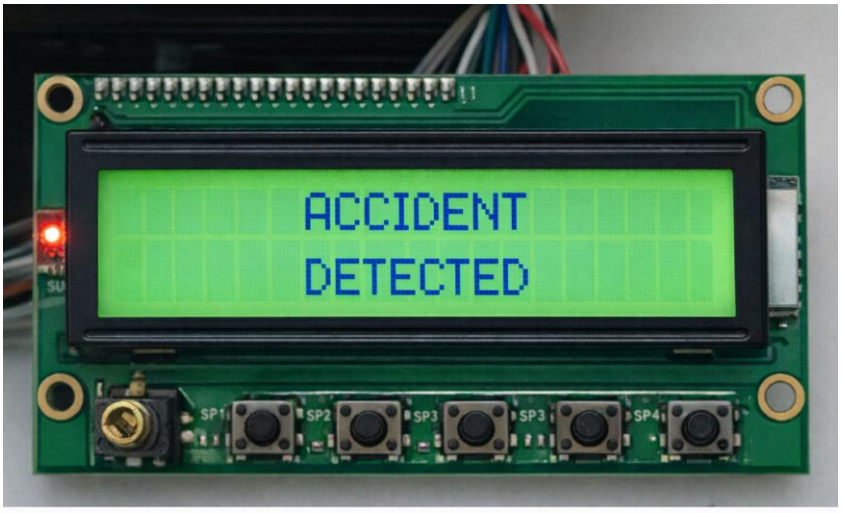


Figure 5.3: Sensor Monitoring Output

Figure 5.3: Accident Detection Output

5.4 Emergency SMS Alert Generation

In the below screen, the GSM module sends an emergency alert message to predefined contact numbers.

The emergency SMS contains information indicating that an accident has occurred and that immediate

assistance is required.

Example SMS message:

"Accident detected! Please provide immediate assistance to the rider."

The SMS alert is sent to:

• Family members

• Emergency contacts

• Medical assistance services

The GSM module communicates with the microcontroller through serial communication using **AT commands** to transmit the message.

This feature ensures quick communication during emergency situations.

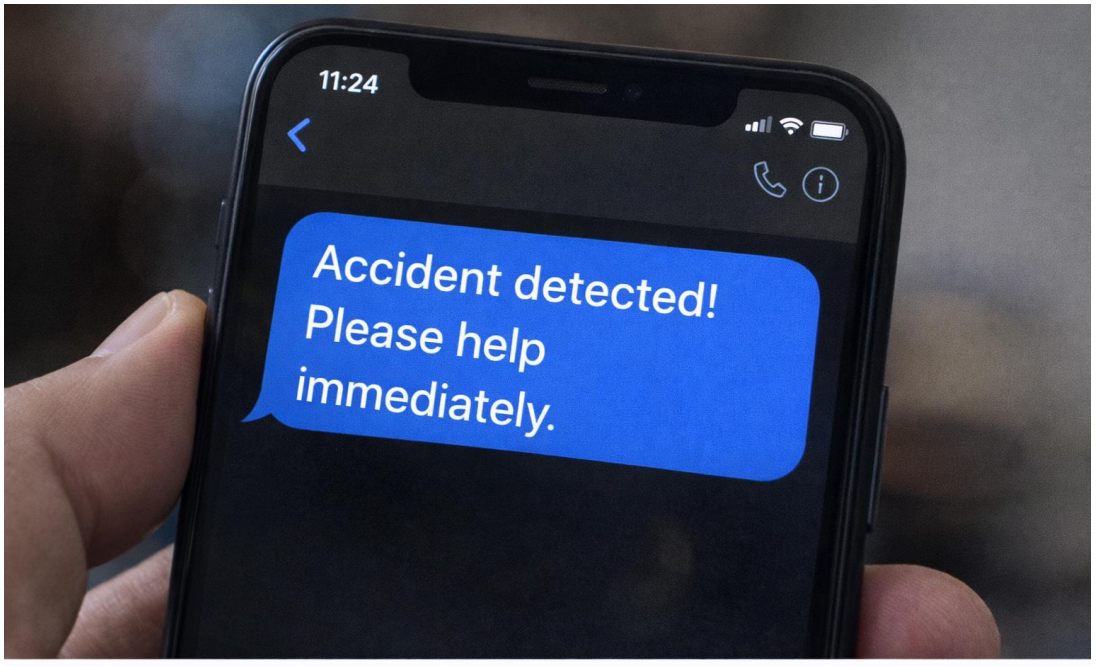


Figure 5.4: Emergency SMS Alert Notification

Figure 5.4: Emergency SMS Alert Notification

5.5 LCD Display Output

In the below screen, the LCD display shows real-time messages generated by the system.

The LCD module provides visual feedback regarding system status and accident alerts. Typical messages displayed include:

• System initialization message
• Monitoring status message
• Accident detection alert
• Emergency notification confirmation

Example display messages:

“SYSTEM READY”
“MONITORING HELMET”
“ACCIDENT DETECTED”
“SMS SENT SUCCESSFULLY”

The LCD display helps users understand the current system status and confirm that emergency alerts have been triggered.

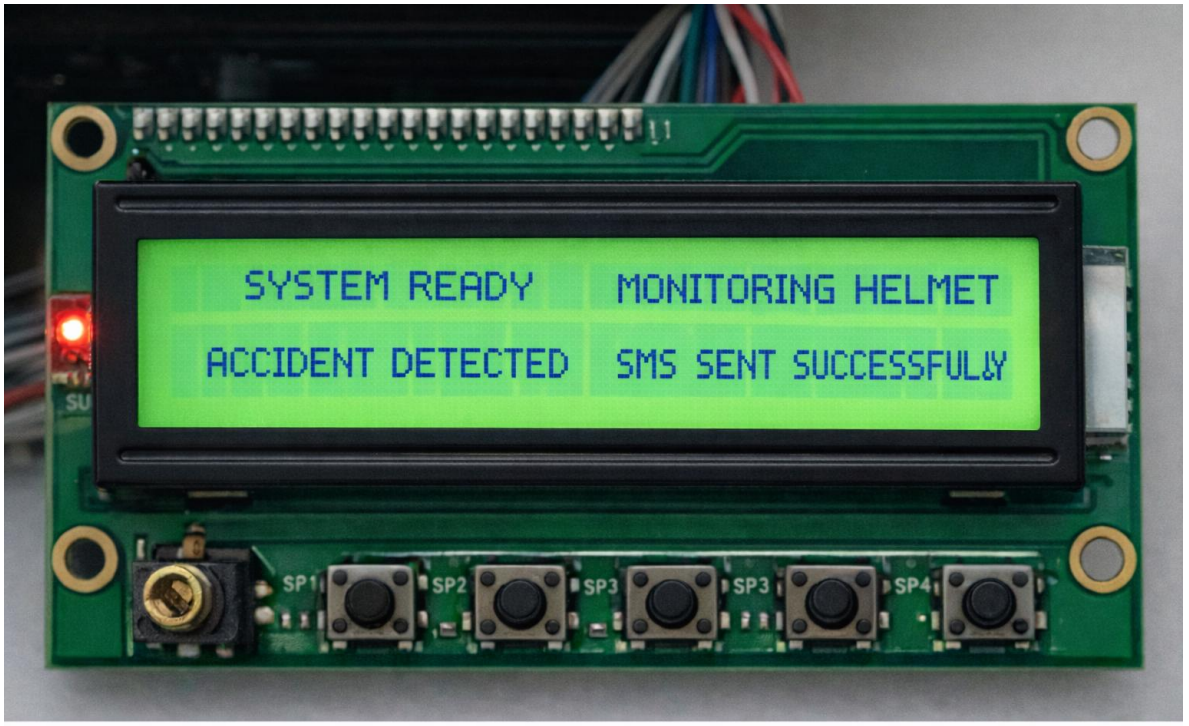


Figure 5.5: LCD Display Output of Smart Helmet System

Figure 5.5: LCD Display Output of Smart Helmet System

5.6 Buzzer Alert Activation

In the below screen, the buzzer alert mechanism is activated when an accident is detected.

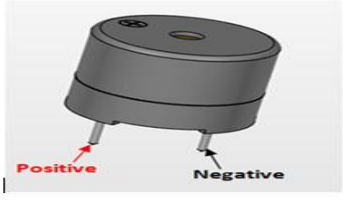
The buzzer generates a loud audible alarm to attract attention from nearby people and indicate that an emergency situation has occurred.

The buzzer alert helps in situations where the rider may be unconscious or unable to request help.

The alert continues for a specific time duration until the system completes the emergency communication process.

This feature enhances the chances of immediate rescue by notifying nearby individuals.

Figure 5.6: Buzzer Alert Activation



6.VALIDATION

6.1 INTRODUCTION

The validation of the **Safety and Smart Helmet System for Bike Riders** is performed through systematic testing to ensure the reliability, accuracy, and proper functioning of all hardware and software components.

The validation process verifies whether the system correctly detects accidents, activates alerts, and sends emergency notifications. Testing is performed under different operating conditions to evaluate system performance and response time.

The validation procedure includes:

• Hardware component testing

• Sensor accuracy testing

• GSM communication testing

• LCD display verification

• Buzzer alert testing

• Integrated system testing

6.1 INTRODUCTION

The Smart Helmet System integrates multiple electronic components such as the **Arduino microcontroller, MEMS sensor, GSM module, LCD display, and buzzer** to provide an intelligent accident detection and alert mechanism.

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Initially, individual components are tested separately to ensure proper operation. After successful component testing, the components are integrated and tested together as a complete system.

The system validation focuses on the following aspects:

• Accurate detection of abnormal motion using the MEMS sensor

• Proper activation of buzzer alerts during emergency situations

• Successful transmission of SMS messages through the GSM module

• Correct display of system messages on the LCD screen

• Reliable communication between the microcontroller and peripheral devices

The system response time is also evaluated to ensure that emergency alerts are generated quickly after accident detection.

Through this validation process, the Smart Helmet System demonstrates reliable

performance.

6.2 TEST CASES

The following test cases were performed to validate the functionality of the Smart Helmet System.

TABLE 6.2.1 SYSTEM INITIALIZATION

Test Case ID	Test Case Name	Purpose	Test Case	Output
1	System Initialization	To verify system startup process	Power ON the Smart Helmet System	LCD displays "SMART HELMET SYSTEM – SYSTEM READY"

TABLE 6.2.2 SENSOR MONITORING

Test Case ID	Test Case Name	Purpose	Test Case	Output
1	Sensor Monitoring	To verify MEMS sensor functionality	Move helmet normally	System monitors motion without triggering alert

TABLE 6.2.3 ACCIDENT DETECTION

Test Case ID	Test Case Name	Purpose	Input	Output
1	Accident Detection Test	To check accident detection capability	Sudden shock or heavy vibration applied to helmet	LCD displays "ACCIDENT DETECTED" and buzzer activates

TABLE 6.2.4 SMS ALERT TEST

Test Case ID	Test Case Name	Purpose	Input	Output
1	SMS Alert Test	To verify GSM communication	Accident detected by sensor	Emergency SMS sent to registered contact number

TABLE 6.2.5 BUZZER ALERT TEST

Test Case ID	Test Case Name	Purpose	Input	Output
1	Buzzer Alert Test	To verify buzzer activation	Accident detection triggered	Buzzer produces loud alert sound

TABLE 6.2.6 LCD DISPLAY TEST

Test Case ID	Test Case Name	Purpose	Input	Output
1	LCD Display Test	To verify system message display	System powered ON	LCD displays system status messages

Validation Summary

The validation results confirm that the Smart Helmet System successfully performs all required functions, including:

• Continuous monitoring of helmet movement

• Accurate detection of accident conditions

• Immediate activation of buzzer alerts

• Successful transmission of emergency SMS notifications

• Real-time display of system status messages

7.CONCLUSION & FUTURE ASPECTS

7.1 CONCLUSION & FUTURE ASPECTS

The **Safety and Smart Helmet System for Bike Riders** project successfully demonstrates the use of embedded systems and sensor technology to enhance the safety of motorbike riders. The system integrates various electronic components such as an Arduino microcontroller, MEMS sensor, GSM module, LCD display, and buzzer to detect accidents and immediately notify emergency contacts.

The project addresses one of the major problems in road safety, which is the delay in providing medical assistance after accidents. By automatically detecting abnormal movements or impacts using the MEMS sensor, the system quickly triggers alerts and sends SMS notifications through the GSM module. This rapid response mechanism helps reduce the time required to inform emergency services and family members.

The system also provides real-time feedback through the LCD display and buzzer alerts, allowing the rider to be aware of the system status. The integration of these components ensures reliable accident detection and communication functionality.

Overall, the Smart Helmet System is a practical and cost-effective solution that can significantly contribute to improving road safety. The project demonstrates how embedded technology and communication systems can be combined to develop intelligent safety devices for real-world applications.

7.1 PROJECT CONCLUSION

The **Safety and Smart Helmet System for Bike Riders** project was successfully designed and implemented to improve the safety of motorbike riders. The system uses sensors and communication modules to detect accidents and send emergency alerts automatically.

The MEMS sensor continuously monitors the movement and orientation of the helmet. When sudden abnormal movement or strong vibration is detected, the system identifies it as a possible accident. The microcontroller then activates the buzzer alert and displays warning messages on the LCD screen.

Simultaneously, the GSM module sends an emergency SMS notification to predefined contact numbers to inform them about the accident. This ensures that help can be arranged quickly without depending solely on the injured rider.

7.2 FUTURE ASPECTS

Although the current Smart Helmet System performs its intended functions effectively, several improvements can be implemented in the future to enhance the system's capabilities and efficiency.

1. GPS Location Tracking

Future versions of the system can include a GPS module to send the exact accident location along with the emergency alert message. This will help emergency responders reach the accident site quickly.

2. Mobile Application Integration

A mobile application can be developed to monitor helmet status, receive alerts, and track accident information in real time.

3. Alcohol Detection System

An alcohol sensor can be added to detect whether the rider has consumed alcohol. If alcohol is detected, the system can prevent the bike from starting.

4. Helmet Wearing Detection

Sensors can be used to detect whether the rider is wearing the helmet properly before starting the bike, improving safety compliance.

5. IoT Cloud Integration

The system can be connected to cloud platforms to store accident data and monitor rider safety statistics over time.

6. Advanced Sensors

More advanced sensors such as gyroscopes and vibration sensors can be integrated to improve accident detection accuracy.

7. Emergency Call Feature

In addition to SMS alerts, the system can automatically make emergency calls to hospitals or emergency services.

By implementing these improvements, the Smart Helmet System can evolve into a more advanced intelligent safety solution capable of providing enhanced protection for riders and reducing road accident fatalities.

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8.2 PROJECT SOURCE / CODE REPOSITO

The source code and implementation files for the **Safety and Smart Helmet System for Bike Riders** project are maintained in a GitHub repository for version control and future development